# Johannes U. Lange

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## RESEARCH INTERESTS

Dark Energy, Gravitational Lensing, Galaxy Formation, Statistical Methods and Machine Learning

## POSITIONS

**American University** 08/2024 – present
Assistant Professor
**University of Michigan** 09/2022 – 07/2024
Leinweber Center for Theoretical Physics Fellow
**Stanford University** 09/2021 – 08/2022
Stanford–Santa Cruz Cosmology Postdoctoral Fellow
**University of California, Santa Cruz** 09/2019 – 08/2021
Stanford–Santa Cruz Cosmology Postdoctoral Fellow

## EDUCATION

**Yale University** 08/2014 – 08/2019
M.Sc., M.Phil, Ph.D. in Astronomy
**Ruprecht-Karls-Universität Heidelberg** 09/2012 – 08/2014
Master of Science in Physics
**Freie Universität Berlin** 10/2009 – 08/2012
Bachelor of Science in Physics

## TEACHING

**Instructor, American University** Fall 2025
Course: Data Mining and Machine Learning for Natural Sciences
**Instructor, American University** Spring 2025, Spring 2026
Course: Changing Views of the Universe
**Instructor, American University** Fall 2024
Course: Modern Physics
**Postdoctoral Course on STEM Teaching, University of Michigan** Winter 2024
The Postdoctoral Short Course on College Teaching in STEM is a comprehensive 10-week program for postdocs to teach effectively as future faculty members.
**Adjunct Lecturer, University of Michigan** Fall 2023
Course: Naked-Eye Astronomy
**Certificate of College Teaching Preparation, Yale University** 2014-2019
The Certificate of College Teaching Preparation (CCTP) is an opportunity for graduate students to engage in a comprehensive training program in effective college teaching.
**Teaching Fellow, Yale University** Spring 2016, Spring 2018
Course: Astrostatistics and Data Mining
**Teaching Fellow, Yale University** Fall 2017, Fall 2014
Course: Introduction to Astronomical Observing

**Teaching Fellow, Yale University** Fall 2015
Course: Introduction to Cosmology
**Teaching Fellow, Yale University** Spring 2015
Course: Gravity, Astrophysics, and Cosmology

## ADVISING

**Simona Sotiri (undergraduate)** 2025-present
Topic: The Lensing-is-Low Problem in FLAMINGO
**Henry Gray (undergraduate)** 2025-present
Topic: The Dependence of Assembly Bias on Galaxy Properties
**Alisun Coldiron (undergraduate)** 2025-present
Topic: Galaxy-Halo Connection in FLAMINGO
**Abigail Fisher (undergraduate)** 2025-present
Topic: Photometric Redshift Calibration
**Alexandra Wells (undergraduate, co-adviser)** 2024-2025
Topic: Cosmology from Non-Linear Scales
**Alexandra Doytcheva (undergraduate)** 2023-2024
Topic: Galaxy Clustering and Control Variates
**Filomela Gerou (undergraduate)** 2022-2024
Topic: Galaxy Clustering and Control Variates
**Gilad Pifko (undergraduate)** 2022-2023
Topic: Relationship between Galaxy and Dark Matter Halo Size
**Simon Wu (undergraduate)** 2022-2023
Topic: Gravitational Lensing Contribution from Subhalos
**Garv Shah (undergraduate)** 2022-2023
Topic: Boosting Importance Nested Sampling with Neural Networks
**Juliana Karp (undergraduate)** 2022-2023
Topic: Anisotropic Satellite Galaxy Quenching
**Diana Blanco (graduate, co-adviser)** 2021-2025
Topic: Photometric Redshift Calibration
**Enia Xhakaj (graduate, co-adviser)** 2019-2023
Topic: Gravitational Lensing

## FIRST-AUTHOR AND STUDENT-LED PAPERS


[17] J. U. Lange et al. "A Unified Photometric Redshift Calibration for Weak Lensing Surveys Using the Dark Energy Spectroscopic Instrument". *ApJ* 998.2, 263 (Feb. 2026).

[16] J. U. Lange et al. "Cosmological Constraints from Full-Scale Clustering and Galaxy-Galaxy Lensing with DESI DR1". *arXiv e-prints*, arXiv:2512.15962 (Dec. 2025).

[15] A. Doytcheva, F. V. Gerou, and J. U. Lange. "High-precision Galaxy Clustering Predictions from Small-volume Hydrodynamical Simulations via Control Variates". *ApJ* 977.2, 184 (Dec. 2024).

[14] J. U. Lange et al. "Systematic Effects in Galaxy-Galaxy Lensing with DESI". *The Open Journal of Astrophysics* 7, 57 (July 2024).

[13] J. U. Lange. "NAUTILUS: boosting Bayesian importance nested sampling with deep learning". *MNRAS* 525.2 (Oct. 2023), pp. 3181–3194.

[12] J. S. M. Karp, J. U. Lange, and R. H. Wechsler. "Anisotropic Satellite Galaxy Quenching: A Unique Signature of Energetic Feedback by Supermassive Black Holes?" *ApJL* 949.1, L13 (May 2023).
[11] J. U. Lange et al. "Constraints on $S_8$ from a full-scale and full-shape analysis of redshift-space clustering and galaxy-galaxy lensing in BOSS". *MNRAS* 520.4 (Apr. 2023), pp. 5373–5393.
[10] J. U. Lange et al. "Five per cent measurements of the growth rate from simulation-based modelling of redshift-space clustering in BOSS LOWZ". *MNRAS* 509.2 (Jan. 2022), pp. 1779–1804.
[9] J. U. Lange et al. "On the halo-mass and radial scale dependence of the lensing is low effect". *MNRAS* 502.2 (Apr. 2021), pp. 2074–2086.
[8] J. U. Lange et al. "Cosmological Evidence Modelling: a new simulation-based approach to constrain cosmology on non-linear scales". *MNRAS* 490.2 (Dec. 2019), pp. 1870–1878.
[7] J. U. Lange et al. "New perspectives on the BOSS small-scale lensing discrepancy for the Planck ΛCDM cosmology". *MNRAS* 488.4 (Oct. 2019), pp. 5771–5787.
[6] J. U. Lange et al. "Updated results on the galaxy-halo connection from satellite kinematics in SDSS". *MNRAS* 487.3 (Aug. 2019), pp. 3112–3129.
[5] J. U. Lange et al. "Maturing satellite kinematics into a competitive probe of the galaxy-halo connection". *MNRAS* 482.4 (Feb. 2019), pp. 4824–4845.
[4] J. U. Lange et al. "Brightest galaxies as halo centre tracers in SDSS DR7". *MNRAS* 473.2 (Jan. 2018), pp. 2830–2851.
[3] J. U. Lange et al. "Evidence for Non-stellar Rest-frame Near-IR Emission Associated with Increased Star Formation in Galaxies at z ~1". *ApJL* 819.1, L4 (Mar. 2016).
[2] J. U. Lange and M.-C. Chu. "Can galactic dark matter substructure contribute to the cosmic gamma-ray anisotropy?" *MNRAS* 447.1 (Feb. 2015), pp. 939–947.
[1] J. Lange and M. Pohl. "The average GeV-band emission from gamma-ray bursts". *A&A* 551, A89 (Mar. 2013).

## OTHER CO-AUTHOR PAPERS

[35] A. Semenaite et al. "Joint cosmological fits to DESI-DR1 full-shape clustering and weak gravitational lensing in configuration space". *The Open Journal of Astrophysics* 9 (May 2026), p. 62334.
[34] DESI Collaboration et al. "Data Release 1 of the Dark Energy Spectroscopic Instrument". *AJ* 171.5, 285 (May 2026).
[33] A. Porredon et al. "DESI-DR1 3 × 2 -pt analysis: consistent cosmology across weak lensing surveys". *The Open Journal of Astrophysics* 9 (Apr. 2026), p. 61342.
[32] N. P. A. P. Emas et al. "Validation of the DESI-DR1 3x2-pt analysis: scale cut and shear ratio tests". *The Open Journal of Astrophysics* 9 (Mar. 2026), p. 59886.
[31] A. Polzin et al. "Picture an Astronomer: Best Practices for Retaining Talent in Astrophysics". *arXiv e-prints*, arXiv:2512.24465 (Dec. 2025).
[30] D. Blanco et al. "The Power of DESI for Photometric Redshift Calibration: A Case Study with KiDS-1000". *arXiv e-prints*, arXiv:2512.15964 (Dec. 2025).
[29] R. Ruggeri et al. "Clustering redshift distribution calibration of weak lensing surveys using the DESI-DR1 spectroscopic dataset". *arXiv e-prints*, arXiv:2512.15963 (Dec. 2025).
[28] K. Mitra et al. "BASILISK IV. No $S_8$ Tension with Satellite Kinematics". *arXiv e-prints*, arXiv:2512.14889 (Dec. 2025).
[27] S. J. Rauhut et al. "Testing gravitational physics by combining DESI DR1 and weak lensing datasets using the E_G estimator". *The Open Journal of Astrophysics* 8, 149 (Oct. 2025).

[26] S. Heydenreich et al. "Lensing Without Borders: Measurements of galaxy-galaxy lensing and projected galaxy clustering in DESI DR1". *arXiv e-prints*, arXiv:2506.21677 (June 2025).
[25] M. Kwiecien et al. "Improving galaxy cluster selection with the outskirt stellar mass of galaxies". *PRD* 111.12, 123524 (June 2025).
[24] C. Blake et al. "The DESI-Lensing Mock Challenge: large-scale cosmological analysis of 3x2-pt statistics". *The Open Journal of Astrophysics* 8, 24 (Mar. 2025).
[23] S. Chen et al. "Analysis of DESI×DES using the Lagrangian effective theory of LSS". *PRD* 110.10, 103518 (Nov. 2024).
[22] K. Mitra, F. C. van den Bosch, and J. U. Lange. "BASILISK II. Improved constraints on the galaxy-halo connection from satellite kinematics in SDSS". *MNRAS* 533.3 (Sept. 2024), pp. 3647–3675.
[21] S. Yuan et al. "Redshift evolution and covariances for joint lensing and clustering studies with DESI Y1". *MNRAS* 533.1 (Sept. 2024), pp. 589–607.
[20] Y. Wang et al. "Measuring the Conditional Luminosity and Stellar Mass Functions of Galaxies by Combining the Dark Energy Spectroscopic Instrument Legacy Imaging Surveys Data Release 9, Survey Validation 3, and Year 1 Data". *ApJ* 971.1, 119 (Aug. 2024).
[19] DESI Collaboration et al. "The Early Data Release of the Dark Energy Spectroscopic Instrument". *AJ* 168.2, 58 (Aug. 2024).
[18] E. Xhakaj et al. "Cluster cosmology without cluster finding". *MNRAS* 530.4 (June 2024), pp. 4203–4218.
[17] DESI Collaboration et al. "Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument". *AJ* 167.2, 62 (Feb. 2024).
[16] B. Hadzhiyska et al. "Synthetic light-cone catalogues of modern redshift and weak lensing surveys waith ABACUSSUMMIT". *MNRAS* 525.3 (Nov. 2023), pp. 4367–4387.
[15] R. Ruggeri et al. "A data compression and optimal galaxy weights scheme for Dark Energy Spectroscopic Instrument and weak lensing data sets". *MNRAS* 525.3 (Nov. 2023), pp. 3865–3878.
[14] K. Wang et al. "Evidence of galaxy assembly bias in SDSS DR7 galaxy samples from count statistics". *MNRAS* 516.3 (Nov. 2022), pp. 4003–4024.
[13] DESI Collaboration et al. "Overview of the Instrumentation for the Dark Energy Spectroscopic Instrument". *AJ* 164.5, 207 (Nov. 2022).
[12] S. Huang et al. "The outer stellar mass of massive galaxies: a simple tracer of halo mass with scatter comparable to richness and reduced projection effects". *MNRAS* 515.4 (Oct. 2022), pp. 4722–4752.
[11] E. Xhakaj et al. "Beyond mass: detecting secondary halo properties with galaxy-galaxy lensing". *MNRAS* 514.2 (Aug. 2022), pp. 2876–2890.
[10] K. Dawson et al. "Snowmass2021 Cosmic Frontier White Paper: High Density Galaxy Clustering in the Regime of Cosmic Acceleration". *arXiv e-prints*, arXiv:2203.07291 (Mar. 2022).
[9] A. Leauthaud et al. "Lensing without borders - I. A blind comparison of the amplitude of galaxy-galaxy lensing between independent imaging surveys". *MNRAS* 510.4 (Mar. 2022), pp. 6150–6189.
[8] K. Wang et al. "Concentrations of dark haloes emerge from their merger histories". *MNRAS* 498.3 (Nov. 2020), pp. 4450–4464.
[7] F. C. van den Bosch, J. U. Lange, and A. R. Zentner. "Basilisk: Bayesian hierarchical inference of the galaxy-halo connection using satellite kinematics - I. Method and validation". *MNRAS* 488.4 (Oct. 2019), pp. 4984–5013.
[6] K. Wang et al. "How to optimally constrain galaxy assembly bias: supplement projected correlation functions with count-in-cells statistics". *MNRAS* 488.3 (Sept. 2019), pp. 3541–3567.

[5] A. R. Zentner et al. "Constraints on assembly bias from galaxy clustering". *MNRAS* 485.1 (May 2019), pp. 1196–1209.
[4] D. Campbell et al. "The galaxy clustering crisis in abundance matching". *MNRAS* 477.1 (June 2018), pp. 359–383.
[3] A. S. Villarreal et al. "The immitigable nature of assembly bias: the impact of halo definition on assembly bias". *MNRAS* 472.1 (Nov. 2017), pp. 1088–1105.
[2] E. J. Nelson et al. "Where Stars Form: Inside-out Growth and Coherent Star Formation from HST H$\alpha$ Maps of 3200 Galaxies across the Main Sequence at $0.7 < z < 1.5$". *ApJ* 828.1, 27 (Sept. 2016).
[1] I. G. Momcheva et al. "The 3D-HST Survey: Hubble Space Telescope WFC3/G141 Grism Spectra, Redshifts, and Emission Line Measurements for ~100,000 Galaxies". *ApJS* 225.2, 27 (Aug. 2016).

## INVITED TALKS

**Particle Theory & Cosmology Seminar** 04/2026
Johns Hopkins University
**Astronomy Seminar** 05/2025
Carnegie EPL
**Cosmology Seminar** 04/2025
University of California, Berkeley
**CTC Seminar Series** 03/2025
University of Maryland
**Physics Colloquium** 04/2024
University of Hawaii
**ITP Cosmology Seminar** 12/2023
Ruprecht-Karls-Universität Heidelberg
**Frontiers of Nested Sampling Workshop** 07/2023
42nd International Workshop on Bayesian Inference and Maximum Entropy Methods in Science and Engineering
**Webinar Series** 06/2023
National Observatory in Rio de Janeiro
**Early Career Researcher Cosmology Seminar** 11/2022
Korea Astronomy and Space Science Institute
**HEAP Seminar** 12/2021
University of Utah
**Astronomy Colloquium** 09/2021
Swinburne University of Technology
**Growth of Structure Webinar** 07/2021
University of California, Santa Cruz
**Growth of Structure Webinar** 06/2021
University of California, Santa Cruz
**Research Progress Meeting** 01/2019
Lawrence Berkeley National Laboratory
**CCAPP Seminar** 01/2019
Center for Cosmology and AstroParticle Physics
**BCCP Seminar** 09/2018
University of California, Berkeley

**The Galaxy-Halo Connection Across Cosmic Time** 07/2017
Kavli Institute for Theoretical Physics

## OUTREACH

**Public Observing Nights at AU, Washington, DC** 2024 - present
**KIPAC Public Lecture, Palo Alto, CA** 07/2022
**Class at Stanford Splash, Palo Alto, CA** 11/2021
**Talk at Astronomy on Tap, New Haven, CT** 06/2019
**Talk at Institute for Learning in Retirement, New Haven, CT** 04/2019
**Talks at Leitner Family Observatory, New Haven, CT** 02/2018, 05/2019
**Talks at Open Labs Science Cafe, Yale University, New Haven, CT** 10/2017, 04/2019
**Member of Open Labs, Yale University, New Haven, CT** 2016 - 2019
**Tutor at New Haven Reads, New Haven, CT** 2015 - 2018
**Member of UCSB Physics Circus, UC Santa Barbara, Santa Barbara, CA** 2012

## HONORS AND AWARDS

**Brouwer Ph.D. Thesis Prize, Yale University**
**Cosmology Fellowship, UC Santa Cruz and Stanford University**
**Graduate Fellowship Program, Kavli Institute for Theoretical Physics**
**Henry A. Smith Fellowship, Yale University**
**DAAD (German Academic Exchange Service) Scholarship**
**Deutschlandstipendium National Scholarship Program**
**Ernst Reuter Scholarship, Free University of Berlin**

## LEADERSHIP ROLES

**Co-chair of the Roman External Synergies Science Focus Group** 2026 - present
**Co-Chair of the NASA Cosmic Structure Science Interest Group** 2025 - present
**Co-Chair of the DESI desilike Topical Group** 2024 - present
**Co-Chair of the DESI $C^3$ Working Group** 2022 - 2024

## COMMITTEE MEMBERSHIP

**Stanford Physics Equity & Inclusion Committee** 2021 - 2022
**DESI Professional Development Mentoring Program** 2021 - present
**DESI Early Career Scientists Committee** 2021 - 2022
**UCSC Astronomy Department Colloquium Committee** 2019 - 2020
**Yale Graduate Admissions Committee** 2018 - 2019

## REVIEWER SERVICE

**Astronomy & Astrophysics**
**Monthly Notices of the Royal Astronomical Society**
**The Astrophysical Journal**

**Journal of Cosmology and Astroparticle Physics**
**Physical Review Letters**
**Discover Space**
**DESI Internal Reviewer**
**National Science Foundation**

## SCIENTIFIC ORGANIZING COMMITTEES

**NASA's Physics of the Cosmos Early Career Workshop** 09/2025
**KICP Workshop** 08/2023
Lensing at Different Scales: Strong, Weak, and Synergies Between the Two
**Michigan Cosmology Summer School 2023** 06/2023
**KITP Online Conference** 08/2020
The Galaxy-Halo Connection Across Cosmic Time: Recent Updates
**KIPAC Online Workshop** 07/2020
Precision Measurements and Modeling of Lensing and Clustering in the DESI Era

## REFERENCES

**Frank C. van den Bosch**
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**Andrew P. Hearin**
Argonne National Laboratory
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**Alexie Leauthaud**
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**Dragan Huterer**
University of Michigan
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